

No Uniform Polynomial Exponent Gap for Positive Kreiss Bounded Semigroups on L^2

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Status: partial result, likely valid; human review requested.

1 Source question

Arnold and Coine ask in Question 3.9(2) of [1] whether every uniformly eventually positive Kreiss bounded C_0 -semigroup $(T_t)_{t \geq 0}$ on an L^p -space, $1 < p < \infty$, satisfies

$$\|T_t\| = O(t^{1-\varepsilon})$$

for some $\varepsilon \in (0, 1)$. The question appears on page 11 of the arXiv PDF and is reproduced below.

- Question 3.9.** (1) In Theorem 3.1, we assumed that the semigroup is uniformly eventually positive and the same assumption was made in [18] to prove the equality between the growth bound and the spectral bound of its generator. It is not known whether this equality holds if we assume that the semigroup is individually eventually positive. Hence, it is natural to ask if Theorem 3.1 remains true if we replace "uniformly eventually positive" by "individually eventually positive".
- (2) In view of Section 4, we can also ask whether the estimate in Theorem 3.1 can be improved. For example, if $(T_t)_{t \geq 0}$ is a uniformly eventually positive Kreiss bounded C_0 -semigroup on L^p , $1 < p < +\infty$, do we have $\|T_t\| = \mathcal{O}(t^{1-\epsilon})$ for some $\epsilon \in (0, 1)$?

We do not settle the quantifier "for each semigroup there exists ε ." We prove a sharpness result: ε cannot be chosen as a function of the ambient exponent p alone, even at $p = 2$ and even for semigroups that are positive from time zero.

2 Proof intuition

Bonilla and Muller constructed, for every prescribed $\delta > 0$, a positive Cesaro-bounded operator T on a Hilbert lattice whose powers are bounded below by a constant times $n^{1-\delta}$ [2, Theorem 2.3]. We turn the discrete dynamics into a continuous semigroup by moving functions around the unit interval and applying one copy of T each time the moving point crosses the endpoint.

At integer times this suspension acts pointwise as T^n , so it preserves the large power norms exactly. The only nonformal step is to show that Cesaro boundedness also survives suspension. Over an integer interval $[0, N]$, the integral of the suspension splits into an initial fragment, the discrete partial sum $I + T + \dots + T^{N-1}$ applied to the integral of the function, and a final fragment multiplied by T^N . The first two are controlled directly, and T^N is the difference

of two consecutive partial sums. Hence the continuous Cesaro means are uniformly bounded. Positivity then lets us invoke the Cesaro–Kreiss equivalence from [1, Proposition 2.6].

3 The suspension lemma

Let X be a Banach space and $T \in \mathcal{L}(X)$. On

$$\mathcal{E} := L^2([0, 1); X)$$

define, for $t \geq 0$ and $0 \leq s < 1$,

$$(S(t)f)(s) := T^{m(t,s)}f(s+t-m(t,s)), \quad m(t,s) := \lfloor s+t \rfloor. \quad (1)$$

Lemma 1 (Basic suspension properties). *The family $(S(t))_{t \geq 0}$ is a C_0 -semigroup on $\mathcal{E}$. If X is a Banach lattice and T is positive, then every $S(t)$ is positive. Moreover, for every $n \in \mathbb{N}_0$,*

$$\|S(n)\| = \|T^n\|. \quad (2)$$

Proof. Writing $s+t = m+r$ with $m \in \mathbb{N}_0$ and $r \in [0, 1)$, and then writing $r+u = m'+r'$, gives

$$m(t+u, s) = m + m', \quad s+t+u - m(t+u, s) = r'.$$

This proves $S(t+u) = S(t)S(u)$. For fixed t , the integer $m(t, s)$ takes at most two values, so $S(t)$ is bounded.

For $0 < t < 1$ one has

$$(S(t)f)(s) = \begin{cases} f(s+t), & 0 \leq s < 1-t, \\ Tf(s+t-1), & 1-t \leq s < 1. \end{cases}$$

Translation continuity in L^2 , together with absolute continuity of the integral on the two boundary intervals of length t , implies $\|S(t)f - f\|_{\mathcal{E}} \rightarrow 0$ as $t \downarrow 0$. The semigroup law yields strong continuity at every time.

Positivity follows immediately from (1). Finally, when $t = n$ is an integer, $m(n, s) = n$ for almost every $s \in [0, 1)$ and

$$(S(n)f)(s) = T^n f(s).$$

The induced pointwise operator on $L^2([0, 1); X)$ has norm $\|T^n\|$: the upper bound is immediate, while the reverse inequality follows by testing constant functions with values in an approximately norming unit vector for T^n . $\square$

Lemma 2 (Cesaro boundedness survives suspension). *Assume*

$$C_T := \sup_{N \geq 1} \frac{1}{N} \left\| \sum_{k=0}^{N-1} T^k \right\| < \infty. \quad (3)$$

Then the suspension (1) is continuously Cesaro bounded:

$$\sup_{\tau > 0} \frac{1}{\tau} \left\| \int_0^\tau S(t) dt \right\| < \infty.$$

Proof. Put $B_N := \sum_{k=0}^{N-1} T^k$. From (3),

$$\|B_N\| \leq C_T N, \quad \|T^N\| = \|B_{N+1} - B_N\| \leq C_T(2N+1). \quad (4)$$

Fix $N \geq 1$ and $f \in \mathcal{E}$. With Bochner integrals in X , set

$$g := \int_0^1 f(v) dv, \quad h_-(s) := \int_s^1 f(v) dv, \quad h_+(s) := \int_0^s f(v) dv.$$

Changing variables from t to $s + t$ and splitting at the integer points gives, for almost every $s \in [0, 1]$,

$$\int_0^N (S(t)f)(s) dt = h_-(s) + (B_N - I)g + T^N h_+(s). \quad (5)$$

Cauchy–Schwarz gives

$$\|g\|_X \leq \|f\|_{\mathcal{E}}, \quad \|h_-\|_{\mathcal{E}} \leq \|f\|_{\mathcal{E}}, \quad \|h_+\|_{\mathcal{E}} \leq \|f\|_{\mathcal{E}}.$$

Consequently, (4) and (5) imply

$$\left\| \int_0^N S(t) dt \right\| \leq C(1 + N)$$

with C depending only on C_T .

For arbitrary $\tau = N + u$ with $N \in \mathbb{N}$ and $0 \leq u < 1$, the remaining integral over $[N, N + u]$ has norm at most

$$\sup_{N \leq t \leq N+1} \|S(t)\| \leq \max\{\|T^N\|, \|T^{N+1}\|\} = O(N)$$

by (4). Division by $\tau \geq N$ gives a uniform bound. The range $0 < \tau < 1$ is controlled by local boundedness of the C_0 -semigroup. $\square$

4 No uniform exponent gap

Theorem 3. *For every $\delta \in (0, 1)$ there are an L^2 -space $\mathcal{E}$ and a positive Kreiss bounded C_0 -semigroup $(S(t))_{t \geq 0}$ on $\mathcal{E}$ such that*

$$\|S(n)\| \geq \frac{1}{3}(n+1)^{1-\delta} \quad (n \in \mathbb{N}_0). \quad (6)$$

Hence there is no number $\varepsilon_2 > 0$ for which every positive Kreiss bounded C_0 -semigroup on every L^2 -space satisfies $\|S(t)\| = O(t^{1-\varepsilon_2})$.

Proof. Fix $\delta \in (0, 1)$. By [2, Theorems 2.2 and 2.3], there are a Hilbert lattice H and a positive Cesaro-bounded weighted shift $T \in \mathcal{L}(H)$ such that

$$\|T^n\| \geq \frac{1}{3}(n+1)^{1-\delta} \quad (n \in \mathbb{N}_0).$$

Their H is a Hilbert direct sum of finite-dimensional coordinate lattices, so H is lattice isometric to an ℓ^2 -space. Therefore $\mathcal{E} = L^2([0, 1]; H)$ is an L^2 -space.

Let $(S(t))$ be the suspension of T . Lemma 1 makes it a positive C_0 -semigroup, and Lemma 2 makes it Cesaro bounded. For an individually eventually positive semigroup on a Banach lattice, continuous Cesaro boundedness is equivalent to Kreiss boundedness by [1, Proposition 2.6]; hence $(S(t))$ is Kreiss bounded. Equation (2) now gives (6).

If a universal $\varepsilon_2 > 0$ existed, choose $0 < \delta < \varepsilon_2$. The universal upper bound at integer times would contradict (6), because $n^{1-\delta}/n^{1-\varepsilon_2} = n^{\varepsilon_2-\delta} \rightarrow \infty$. $\square$

Remark 4 (Exact scope). *Theorem 3 does not disprove the literal source question. For the semigroup associated with a fixed δ , it only forces every possible polynomial gap ε to satisfy $\varepsilon \leq \delta$. It remains open here whether each individual semigroup has some positive gap depending on that semigroup.*

5 Verification, limitations, and novelty

No computation is used in the proof. The two identities most relevant for verification are (5), which transfers discrete Cesaro boundedness to continuous time, and (2), which transfers the power lower bound without loss. The former can also be checked directly by partitioning the interval $[s, N + s]$ at its integer points.

A bounded literature check through 22 July 2026 covered [1, 2, 3, 4, 5] and searches for “positive Cesaro bounded power growth”, “positive Kreiss bounded semigroup sublinear polynomial”, and suspension variants. No explicit statement of Theorem 3 was found. Novelty confidence is moderate-low: the discrete example is known and the suspension is standard, while the exact sharpness consequence for Question 3.9(2) appears not to have been recorded in the checked sources.

The recommended human-review focus is Lemma 2 and the invocation of [1, Proposition 2.6]. Once those are accepted, the conclusion is immediate from the Bonilla–Muller lower bound.

References

- [1] L. Arnold and C. Coine, *Growth rate of eventually positive Kreiss bounded C_0 -semigroups on L^p and $\mathcal{C}(K)$* , J. Evol. Equ. 23 (2023), article 7; [arXiv:2207.08443](#).
- [2] A. Bonilla and V. Muller, *Kreiss bounded and uniformly Kreiss bounded operators*, [arXiv:1912.07931](#).
- [3] G. Cohen, C. Cuny, T. Eisner, and M. Lin, *Resolvent conditions and growth of powers of operators*, J. Math. Anal. Appl. 487 (2020), 124035; [arXiv:1912.10507](#).
- [4] L. Arnold and C. Cuny, *On the growth rate of powers of a strongly Kreiss bounded operator on an L^p -space*, [arXiv:2302.14135](#).
- [5] S. Arora, *Eventually positive semigroups: spectral and asymptotic analysis*, Semigroup Forum 110 (2025), 263–295; [arXiv:2405.16371](#).